

- a) True - they are exactly equal b) False - floating-point binary precision issues c) Error - cannot compare floats with == d) True - Python rounds the result

TOPIC 2: STRINGS & REGULAR EXPRESSIONS (Q26–Q50)

Q26. What does "PYTHON"[::-1] return?

- a) "NYTHON" b) "NOHTYP" c) "PYTHNO" d) "PYTHON"

Q27. What is the output of "hello world".split()?

- a) ["hello", "world"] b) Individual characters in a list c) ("hello", "world") d) ["hello world"]

Q28. What does "abc".upper().lower() return?

- a) "ABC" b) "abc" c) "Abc" d) Error

Q29. What is the result of " data ".strip()?

- a) "data" b) " data" c) "data " d) " data "

Q30. What does "data analytics".find("xyz") return?

- a) None b) -1 c) 0 d) False

Q31. What does ",".join(["a", "b", "c"]) return?

- a) "a, b, c" b) "a,b,c" c) ("a","b","c") d) ["a,b,c"]

Q32. What is "Python"[1:4]?

- a) "Pyt" b) "yth" c) "ytho" d) "hon"

Q33. Which string method returns the count of non-overlapping occurrences?

- a) .find() b) .index() c) .count() d) .search()

Q34. What does "Hello World".replace("l", "L", 2) return?

- a) "HeLLo World" b) "HeLLo WorLd" c) "HeLlo World" d) "Hello World"

Q35. What does "hello".index("z") raise?

- a) Returns -1 b) Returns None c) ValueError d) IndexError

Q36. What does f"{42:05d}" produce?

- a) "42000" b) "00042" c) " 0042" d) "42 "

Q37. What does "ATCG" * 3 produce?

- a) "ATCGATCGATCG" b) ["ATCG","ATCG","ATCG"] c) 12 d) Error

Q38. What is len("Hello\n")?

a) 5 b) 6 c) 7 d) Error

Q39. What does `r"\n"` contain?

a) A newline character b) Two chars: backslash + n c) An empty string d) Error

Q40. What does `"Python".startswith(("py","Py"))` return?

a) Error - tuple not allowed b) True c) False d) [False, True]

Q41. What does `"abcabc".replace("b","B")` return?

a) "aBcabc" b) "aBcaBc" c) "abcaBc" d) "aBc"

Q42. What does `re.findall(r"\d+", "a1b22c333")` return?

a) ["1", "22", "333"] b) "1", "22", "333" (separate values) c) ["1", "2", "2", "3", "3", "3"] d) ["1", "2", "3"]

Q43. What does `re.search(r"\bword\b", "keyword")` return?

a) A match object b) None c) "word" d) Error

Q44. What does `"hello"[10]` raise?

a) None b) "" c) IndexError d) " "

Q45. What does `"a:b:c:d".split(":", 2)` return?

a) ["a", "b", "c", "d"] b) ["a", "b", "c:d"] c) ["a:b:c:d"] d) ["a", "b:c:d"]

Q46. Which regex pattern matches ONE OR MORE digits?

a) \d b) \d+ c) \d* d) \D+

Q47. What does `" ".strip()` return?

a) " " b) "" c) None d) " "

Q48. What does `re.sub(r"[aeiou]", "*", "hello")` return?

a) "h*ll0" b) "h*ll*" c) "h***o" d) "*ello"

Q49. What does `"abcdef"[:2]` return?

a) "ace" b) "bdf" c) "abc" d) "def"

Q50. When should you use `re.compile()` over calling `re.search()` directly?

a) Always - compile is always faster b) When the same pattern is used many times in a loop c) Never
- they are identical d) Only when using capture groups

TOPIC 3: LISTS, TUPLES & COMPREHENSIONS (Q51–Q75)

Q51. What does the following print?

```
a = [1,2,3]; b = a; b.append(4); print(a)
```

14	D	isinstance() supports tuple of types. type(x) in [int,float] also works. Never use type(x)==float or int - the or int always evaluates truthy.
15	A	.2f means 2 decimal places, so 3.14159 rounds to 3.14.
16	B	not 0 = True (0 is falsy), then not True = False.
17	C	bool is a subclass of int: True=1, False=0. So 1+1+0 = 2 (integer).
18	B	Python integers have arbitrary precision - 2**1000 works without overflow.
19	B	id() returns the memory address. Equal ids mean same object (identity, not equality).
20	B	PEMDAS in Python: ** > * // % > + - > comparisons. Remember: 2+3*4 = 14, not 20.
21	C	int(None) raises TypeError. int(True)=1 and int(False)=0 are valid, but None cannot be converted.
22	C	You cannot concatenate str and int. Use int("5") + 5 or "5" + str(5).
23	A	divmod(a,b) returns (a//b, a%b): quotient and remainder. 13//4=3, 13%4=1.
24	A	/ always returns float in Python 3. // returns int (if both operands are int). ** returns int.
25	B	0.1+0.2 in binary ≈ 0.30000000000000004. Always use math.isclose() for float comparisons.
26	B	[::-1] reverses the string. P=0,Y=1,T=2,H=3,O=4,N=5 reversed = NOHTYP.
27	A	split() with no argument splits on any whitespace and removes extra

spaces.

28	B	Strings are immutable. Each method returns a new string: <code>.upper()="ABC"</code> , <code>.lower()="abc"</code> .
29	A	<code>strip()</code> removes leading and trailing whitespace. <code>lstrip()</code> and <code>rstrip()</code> do one side each.
30	B	<code>find()</code> returns -1 when the substring is not found. <code>index()</code> raises <code>ValueError</code> instead.
31	B	<code>join()</code> concatenates iterable elements with the separator. No trailing separator.
32	B	Indices: <code>P=0,y=1,t=2,h=3,o=4,n=5</code> . <code>[1:4]</code> means <code>start=1 (y)</code> , <code>stop=4 (exclusive h)</code> = <code>"yth"</code> .
33	C	<code>"hello".count("l")</code> returns 2. <code>.search()</code> does not exist on <code>str</code> - it's from the <code>re</code> module.
34	A	The third argument limits replacements. 2 replaces only the first 2 occurrences of <code>"l"</code> .
35	C	<code>index()</code> raises <code>ValueError</code> for missing substrings. <code>find()</code> is the safe alternative (returns -1).
36	B	<code>05d</code> = integer format, total width 5, zero-padded. 42 needs 3 zeros: <code>"00042"</code> .
37	A	String repetition returns a single repeated string, not a list.
38	B	<code>\n</code> is a single newline character. <code>H-e-l-l-o</code> = 5 chars + <code>\n</code> = 6 total.
39	B	Raw strings (<code>r""</code>) treat backslashes literally. <code>r"\n"</code> contains <code>\</code> and <code>n</code> , not a newline.
40	B	<code>startswith()</code> accepts a tuple of prefixes and returns <code>True</code> if the string starts

		with any of them.
41	B	Without a count limit, replace() replaces ALL occurrences.
42	A	\d+ matches one or more consecutive digits as a group. findall returns all non-overlapping matches.
43	B	\b is a word boundary. In "keyword", "word" is inside another word so boundaries don't match.
44	C	Strings use index-based access. Index 10 on a 5-char string raises IndexError.
45	B	maxsplit=2 means split at most twice, producing 3 parts. The remaining string stays unsplit.
46	B	\d matches exactly one digit. + means "one or more". * means "zero or more". \D matches non-digits.
47	B	strip() removes all whitespace. A string of only whitespace becomes an empty string.
48	B	[aeiou] matches any vowel. In "hello": e and o are vowels → h*I*. h and l are consonants.
49	A	[::2] takes every other character starting from index 0: a(0), c(2), e(4) = "ace".
50	B	re.compile() pre-compiles the pattern into a regex object, avoiding recompilation on every loop iteration.
51	B	b = a creates an alias - both names point to the same list object. Appending via b also changes a.
52	B	Slicing creates a shallow copy. For nested lists, nested objects are still shared.
53	C	pop() with no argument removes and returns the LAST element. pop(0)